

Geomatrix Project File Inventory

Project Overview

Geomatrix is a full-stack land acquisition risk intelligence system. It combines a Next.js frontend with a FastAPI backe

Root Files

package.json

Defines frontend dependencies and scripts for dev/build/start.

next.config.ts

Configures Next.js routes and backend proxy rewrites.

tsconfig.json

TypeScript compiler and module alias settings.

eslint.config.mjs

Linting configuration for frontend code quality.

vercel.json

Deployment configuration for Vercel hosting.

README.md

Main project overview, architecture, setup, and usage instructions.

README_NEW.md

Supplementary updated instructions and project notes.

tb_information_summary.txt

Technical summary for TB/model review including the no-fabrication policy.

all_prompts.txt

Prompt and task information used during project work.

scripts_seed.txt

Seed script-related notes or commands.

next-env.d.ts

Next.js TypeScript environment declarations.

Frontend Structure

app/layout.tsx

Top-level layout wrapper for the app shell.

app/globals.css

Global styling and design system for the interface.

app/page.tsx

Landing or default app entry page.

app/login/page.tsx

Google login / authentication page.

app/dashboard/page.tsx

Executive dashboard with KPIs and summaries.

app/projects/page.tsx

Project master list and filtering UI.

app/projects/[id]/page.tsx

Detailed project intelligence, risk model, SHAP, and explanations.

app/map/page.tsx

Spatial GIS page with project markers and map overlays.

app/alerts/page.tsx

Alert list and acknowledgement interface.

app/data/page.tsx

Data ingestion and upload control panel.

app/model/page.tsx

Model status and training UI.

app/reports/page.tsx

Report generation and export interface.

app/analytics/page.tsx

Portfolio analytics and performance views.

app/settings/page.tsx

System settings and configuration interface.

app/admin/model/page.tsx

Model administration page.

app/api/alerts/route.ts

Alerts API route for the frontend.

app/api/analytics/overview/route.ts

Analytics overview API endpoint.

app/api/analytics/drivers/route.ts

Driver-level analytics API.

app/api/analytics/stages/route.ts

Risk-by-stage analytics API.

app/api/auth/google/route.ts

Google auth API handler.

app/api/documents/upload/route.ts

Document upload endpoint.

app/api/gemini/explain/route.ts

Gemini explanatory proxy route.

app/api/map/geojson/...

GeoJSON endpoints for map data.

app/api/map/projects/...

Project map data API routes.

app/api/predict/route.ts

Prediction route for backend model calls.

app/api/projects/route.ts

Project CRUD API routes.

app/api/projects/[id]/...

Single-project API routes.

app/api/reports/route.ts

Reporting generation API.

Shared Frontend Libraries

lib/api.ts

Shared API helper definitions.

lib/apiClient.ts

Typed client for backend REST calls and business logic requests.

lib/data.ts

Project types, enums, and risk metadata constants.

lib/exportUtils.ts

Utilities for CSV/Excel/PDF export.

lib/reports.ts

Report-building and summary formatting helpers.

components/Shell.tsx

Main layout shell with sidebar and navigation.

components/ExportDropdown.tsx

Export selection control for files.

Backend Structure

geomatrix_v2/main.py

Initializes FastAPI and includes all routers.

geomatrix_v2/database.py

Defines SQLite connection and migration behavior.

geomatrix_v2/models.py

SQLAlchemy models for projects, alerts, predictions, model runs, and ingestion logs.

geomatrix_v2/schemas.py

Pydantic request/response schemas.

geomatrix_v2/seed_db.py

Seeds demo/real-like inventory and initializes model data.

geomatrix_v2/requirements.txt

Python dependencies for backend and ML stack.

Backend Routers

geomatrix_v2/routers/projects.py

Project CRUD, prediction endpoint, and explanation retrieval.

geomatrix_v2/routers/ml.py

Model status and training routes.

geomatrix_v2/routers/ingest.py

CSV/JSON ingestion and validation logic.

geomatrix_v2/routers/alerts.py

Alert listing and acknowledgement operations.

geomatrix_v2/routers/map.py

GeoJSON generation for map views.

geomatrix_v2/routers/gemini.py

Gemini prompt generation and explanation API.

geomatrix_v2/routers/auth.py

Authentication flow and session routes.

ML Pipeline

geomatrix_v2/ml/features.py

Feature extraction and canonical project-to-feature transformation.

geomatrix_v2/ml/train.py

Real labeled training, validation, and artifact writing for the model.

geomatrix_v2/ml/predict.py

Inference engine to run risk prediction using the active model.

geomatrix_v2/ml/explain.py

SHAP-based explanation generation.

geomatrix_v2/ml/evaluate.py

Model performance and scoring helpers.

geomatrix_v2/ml/__init__.py

Package marker for ML module.

Data and Artifacts

geomatrix_v2/sample_data/

Sample or reference data used for development and testing.

geomatrix_v2/uploads/

Uploaded files and ingestion staging files.

geomatrix_v2/static/css/style.css

Static CSS for the older/integration web layer.

geomatrix_v2/static/js/app.js

Static JS used by legacy views.

geomatrix_v2/templates/partials/

Template partials for earlier backend rendering.

geomatrix_v2/model_artifacts/

Serialized model metadata, binaries, and active model pointer files.

Tests

geomatrix_v2/tests/test_ml_pipeline.py

Regression checks for honest model training and risk behavior.

geomatrix_v2/tests/test_real_pipeline_api.py

End-to-end tests covering upload, training, prediction, and SHAP persistence.

Summary of Responsibilities

Frontend handles user experience and reporting.

Backend manages persistence, API logic, ingestion, and ML orchestration.

ML layer ensures only valid real-labeled data is used for risk modeling.

Gemini acts as a recommendation and explanation layer, not the main prediction engine.

SQLite stores the project records, training data, and model metadata.

Final Notes

This repository keeps the UI stable while enforcing integrity: real data only, no fake predictions, no synthetic labels,